

String (10 Problems)

1. Input and print a string

```
#include <iostream>
#include <string>
using namespace std;

// Input and print a string
int main(){
    string first_name,last_name;
    cout <<"Enter First name: ";
    cin >> first_name;

    cin.ignore();

    cout <<"Enter last_name: ";
    cin >>last_name;

    cout <<first_name <<" " <<last_name <<endl;

    return 0;
}
```

2. Find the length of a string

```
#include <iostream>
#include <string>
using namespace std;

//Find the length of a string
int main(){
    string name;
    cout <<"Enter name: ";
    getline(cin,name);

    int count = 0;

    for (char ch : name) {
        if (ch != ' ') {
            count++;
        }
    }

    cout << "Length of String (without spaces): " << count << endl;

    return 0;
}
```

```
}
```

3. Count vowels in a string

```
#include <iostream>
#include<string>
using namespace std;

// Count vowels in a string
int main(){
    string str;
    cout <<"Enter string: ";
    getline(cin,str);

    int count = 0;

    for(char ch:str){
        if (ch == 'a' || ch == 'e' || ch == 'i' || ch == 'o' || ch == 'u' || ch
== 'A' || ch == 'E' || ch == 'I' || ch == 'O' || ch == 'U'){
            count++;
        }
    }
    cout <<count <<" Vowel present is " <<str <<"." <<endl;

    return 0;
}
```

4. Count consonants in a string

```
#include <iostream>
#include <string>
using namespace std;

// Count consonants in a string

int main(){
    string str;
    cout <<"Enter a String: ";
    getline(cin,str);

    int count = 0;

    for(char ch : str){
        if ((ch >= 'A' && ch <= 'Z') || (ch >= 'a' && ch <= 'z')) {
            if (ch == 'a' || ch == 'e' || ch == 'i' || ch == 'o' || ch == 'u'
|| ch == 'A' || ch == 'E' || ch == 'I' || ch == 'O' || ch == 'U'){
                count++;
            }
        }
    }
    cout <<count <<" Consonant present is " <<str <<"." <<endl;

    return 0;
}
```

```
                continue;
            }
            else{
                count++;
            }
        }
    }
    cout <<count;

    return 0;
}
```

5. Count spaces in a string

```
#include <iostream>
#include <string>
using namespace std;

//Count spaces in a string

int main(){
    string str;
    cout <<"Enter a string: ";
    getline(cin,str);

    int count = 0;

    for(char ch : str ){
        if(ch == ' '){
            count++;
        }
    }
    cout <<"Present Space: "<<count <<endl;

    return 0;
}
```

6.Reverse a string

```
#include <iostream>
#include <string>
using namespace std;

int main(){
    string str;
    cout <<"Enter your String: ";
    getline(cin,str);
```

```
for(int i = str.size() - 1; i >= 0 ; i--){
    cout <<str.at(i) <<" ";
}

return 0 ;
}
```

7.Check palindrome string

```
#include <iostream>
#include <string>
using namespace std;

int main() {
    string str;

    cout << "Enter a String: ";
    cin >> str;

    int left = 0;
    int right = str.size() - 1;

    bool isPalindrome = true;

    while (left < right) {
        if (str[left] != str[right]) {
            isPalindrome = false;
            break;
        }
        left++;
        right--;
    }

    if (isPalindrome)
        cout << "Palindrome";
    else
        cout << "Not Palindrome";

    return 0;
}
```

8.Convert string to uppercase

```
#include <iostream>
#include <string>
using namespace std;
```

```
int main()
{
    string str;

    cout << "Enter a String: ";
    getline(cin, str);

    for (int i = 0; i < str.size(); i++)
    {
        if (str[i] >= 'a' && str[i] <= 'z')
        {
            str[i] = str[i] - ('a' - 'A'); // or str[i] -= 32;
        }
    }

    cout << "Uppercase: " << str;

    // with toupper() method using <cctype>
    #include <iostream>
    #include <string>
    #include <cctype>
    using namespace std;

    int main()
    {
        string str;

        cout << "Enter a String: ";
        getline(cin, str);

        for (int i = 0; i < str.size(); i++)
        {
            str[i] = toupper(str[i]);
        }

        cout << "Uppercase String: " << str;

        return 0;
    }

    return 0;
}
```

9.Convert string to lowercase

```
#include <iostream>
#include <string>
#include <cctype>
using namespace std;
```

```
int main() {
    string str;

    cout << "Enter a String: ";
    getline(cin, str);

    for (int i = 0; i < str.size(); i++) {
        str[i] = tolower(str[i]);
    }

    cout << "Lowercase: " << str;

    return 0;
}
```

10.Count occurrences of a character

```
#include <iostream>
#include <string>
using namespace std;

int main() {
    string str;
    char ch;

    cout << "Enter a String: ";
    getline(cin, str);

    cout << "Enter Character: ";
    cin >> ch;

    int count = 0;

    for (int i = 0; i < str.size(); i++) {
        if (str[i] == ch) {
            count++;
        }
    }

    cout << "\"" << ch << " occurs " << count << " times.";

    return 0;
}
```